

# Technical Document #143: Blockchain - Comprehensive Overview

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Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

### Key Takeaways:

- Tokenomics studies the economic systems of cryptocurrency tokens.
- Blockchain is a distributed ledger technology that records transactions across many computers.
- Proof of Stake (PoS) selects validators based on their stake in the network.